IRIS Project Entry and Video Script

Project title

Physical-Only Proprioceptive Recognition of Fifteen Classes with a LEAP Hand

Category

Robotics and Intelligent Machines (ROBO)

Single project guide

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Project abstract

People can often distinguish objects by how their fingers close around them. This study asks whether a robotic hand can make a limited version of that judgment using its own joint positions, without camera images or dedicated tactile arrays. A physical LEAP Hand with four fingers and 16 actuated joints performed the same sequence of three controlled grips on 14 household objects and an empty-hand condition. Motor-current thresholds stopped fingers on resistance, but recognition used only the 16 final joint angles from each grip, giving 48 measurements per trial. Five physical trials per label produced 75 trials across 15 labels.

A small neural classifier was evaluated by three-fold within-dataset validation. Its highest-probability label was correct for 70 of 75 held-out trials (93.3%). A decision rule accepted 67 trials as known classes, of which 65 were correct (97.0% of accepted decisions); eight were uncertain. A nearest-trial method classified 69 of 75 correctly in a separate leave-one-trial-out evaluation. An exploratory 20-label extension, which contains the same 75 trials plus 25 more, gave 85 of 100 correct raw neural predictions. A further exploratory nearest-trial check found 72 of 75 correct using only the first grip, so the benefit of combining three grips is not established by these data.

The findings show that repeatable physical grasp positions contain useful information about the enrolled items. The 15-label set was selected after preliminary analysis, and the validation folds also guided model training and decision thresholds. These are internal, post-selection estimates; performance across later sessions, new object instances, and genuinely unseen classes remains unmeasured.

90 Second Project Video Script

Only the student researcher appears on camera. Fill in the grade before recording.

0–13 seconds

Visual Student on camera with the robotic hand visible

Spoken I'm G. Sai Pranav, a grade [grade] student. My project asks a simple question: can a robotic hand tell what it is holding from where its fingers stop?

13–34 seconds

Visual Show three short grip clips; overlay 16 + 16 + 16 = 48 angles

Spoken I programmed a four-finger LEAP Hand to make three controlled grips. Motor current signals when each finger meets resistance. The recognition model receives only the final joint angles: 48 numbers per trial, with no camera image or tactile sensor array.

34–55 seconds

Visual Show the objects, then the empty hand; display 14 objects + empty hand = 15 labels

Spoken I recorded five physical trials for each of 14 household objects and an empty-hand condition. Across the 75 trials, a small neural network's top label was correct 70 times in within-dataset validation.

55–73 seconds

Visual Show 70/75 and 65/67 accepted, with eight uncertain

Spoken A confidence rule accepted 67 known-class decisions and got 65 right. It marked eight trials uncertain. A simpler nearest-trial comparison got 69 of 75 right in a separate evaluation.

73–90 seconds

Visual Return to camera; show 20 labels: 85/100 and a brief next-test card

Spoken I extended the same dataset to 20 labels, with 85 of 100 raw predictions correct. These results show useful information in finger positions. My next step is testing new sessions and new copies of the objects.

Rehearse at a natural pace and keep the recorded video within 90 seconds.